PETadex: A comprehensive database of plastic-degrading enzyme sequence diversity

Dennis Zhu[1], Thomas Quigley[1], Artem Babaian[1]

[1]Donnelly Centre for Cellular and Biomolecular Research, University of Toronto, Toronto, ON, Canada.

Contributing authors: a.babaian@utoronto.ca;

Abstract

Introduction

Plastics are a cornerstone of modern life, improving the accessibility and affordability of necessities such as medical supplies and food packaging. The same traits responsible for these benefits, low production cost and durability, have also made plastics a persistent, large-scale pollutant: over 400 million tonnes are produced annually, and 13 billion tonnes have accumulated since the mid-20th century [1, 2]. Only a minority of this plastic has been recycled or incinerated; the remainder persists in landfills and the environment, and the carbon it holds exceeds that of all animals on Earth [1–3] (Fig. 1A).

Plastic waste degrades into micro- and nanoplastics (MNPs), which permeate Earth's ecosystems, enter food chains and bioaccumulate in humans [4].

Despite the relative novelty of plastics, some microorganisms have adapted to break them down using plastic-degrading enzymes (PDEs). In 2016, Yoshida *et al.* isolated *Ideonella sakaiensis*, a bacterium that uses polyethylene terephthalate (PET) as its major carbon and energy source, and identified the enzyme responsible, IsPETase [5]. PDEs such as IsPETase represent a promising solution to plastic pollution: they break plastics down into their constituent monomers, which can then be repolymerized into new plastic products or repurposed into biomass (Fig. 1B).

Existing databases of PDEs are limited in scope and diversity. The plastics-active enzymes database (PAZy) contains 495 functionally verified plastic-active enzymes [6], most of them homologs of leaf-branch compost cutinase (LCC) [7] or IsPETase. Sequence-based mining is a well-established method for discovering new enzymes: Danso *et al.* (2018) [8] identified 504 candidates by searching UniProtKB and NCBI-nr, and a further 349 by mining 133 metagenomes. More recently, Erickson *et al.* (2022) [9] screened 87 hot-spring metagenomes (∼38 million sequences), characterizing 74 putative enzymes, testing 51 for activity and identifying 23 novel plastic-active enzymes. These surveys have searched only a small fraction of the sequencing data now publicly available.

In previous work, we built Logan [10], an assembly of the entire Sequence Read Archive (SRA), 27.3 million metagenomic datasets, and searched it with 213 characterized PDEs, retrieving 215.7 million putative PDEs [11]. We demonstrated that novel PDEs can be discovered in this dataset through operationally random selection. However, the lack of annotations prevents strategic selection of candidates for validation.

Here we present PETadex, a methodological and quantitative rebuild of our previous dataset

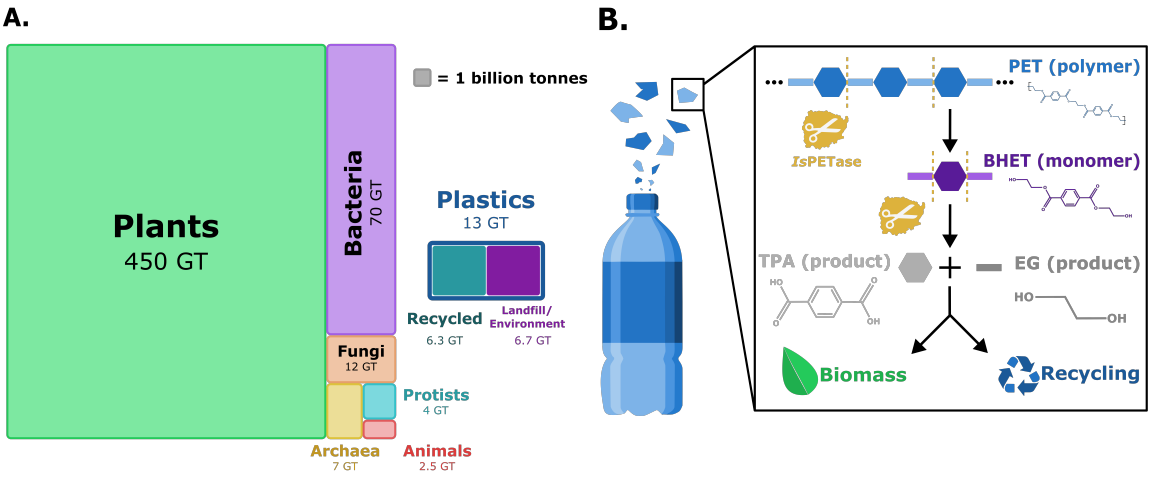


Fig. 1 Plastic waste and its enzymatic degradation. (A) Mass of plastics compared with Earth's major groups of life (1 square = 1 billion tonnes). Of 13 Gt of plastics, 6.3 Gt has been recycled or incinerated and 6.7 Gt remains in landfill or the environment, 2.6-fold the mass of all animals (2.5 Gt). **(B)** Plastic-degrading enzymes as a route out of the plastics crisis: IsPETase hydrolyses PET to the monomer BHET, then to TPA and EG, which feed biomass or recycling.

that supports the targeted selection of PDEs. Using 411 representative PDEs as input, we searched NCBI-nr and the SRA (via the Logan assembly), filtering hits with catalytic-residue-constrained profile HMMs. The resulting dataset of 1.75 billion putative enzymes carries catalytic domain boundaries and is annotated with our full suite of metadata features, ranging from fold assignments, predicted protein structures and the temperature/pH of the biosamples of origin. Candidates can therefore be filtered by catalytic integrity, structural fold and environmental metadata before testing, enabling high-throughput targeted selection using automated machine learning. The enzymes, annotations and a toolkit of methods are freely available at https://petadex.net.

Results

Sequence and structure-based annotation of 609 known plastic-degrading enzymes

Following established workflows for the homology-based retrieval and expansion of a given set of enzymes of interest [8–10], we built a workflow for the retrieval of PDEs from NCBI-nr and the SRA (using the Logan assembly).

Improving upon our previous PETadex dataset, by updating the input search query and fixing several methodological limitations, we present the updated PETadex v2 dataset.

To generate this updated PETadex-260701 dataset, we created a starting search query from combining two versions of the PAZy database, the first from the original PETadex generation (pazy-240826; $n = 212$; August 26th, 2024), and a recently updated version (pazy-260701; $n = 473$; retrieved June 30th, 2026). Sourcing 136 enzymes from pazy-240826, 397 enzymes from pazy-260701 and 76 merged between both, generating a 609 enzyme union set (plasticome). Plasticome was then clustered at 90% amino acid identity to 411 cluster centroids (Fig. 2A). Structure- and sequence-based domain assignment of predicted structures resolved the proteins into 15 distinct protein domains, and all-vs-all alignment further groups the centroids into 46 graph components (Fig. 2A; Methods).

The Plasticome spans several unrelated and unalignable folds, making alignment into a single family multiple sequence alignment unrealiable. Instead we represent the 411 centroids as 56 distinct families based on sequence-similarity, defined as the connected components of a 30% identity and 50% coverage similarity netwrok of all 609 enzymes. These families are distinct from the 46 graph components described above, which use a different threshold and only the 411 centroids. These families are then assigned to 16 fold

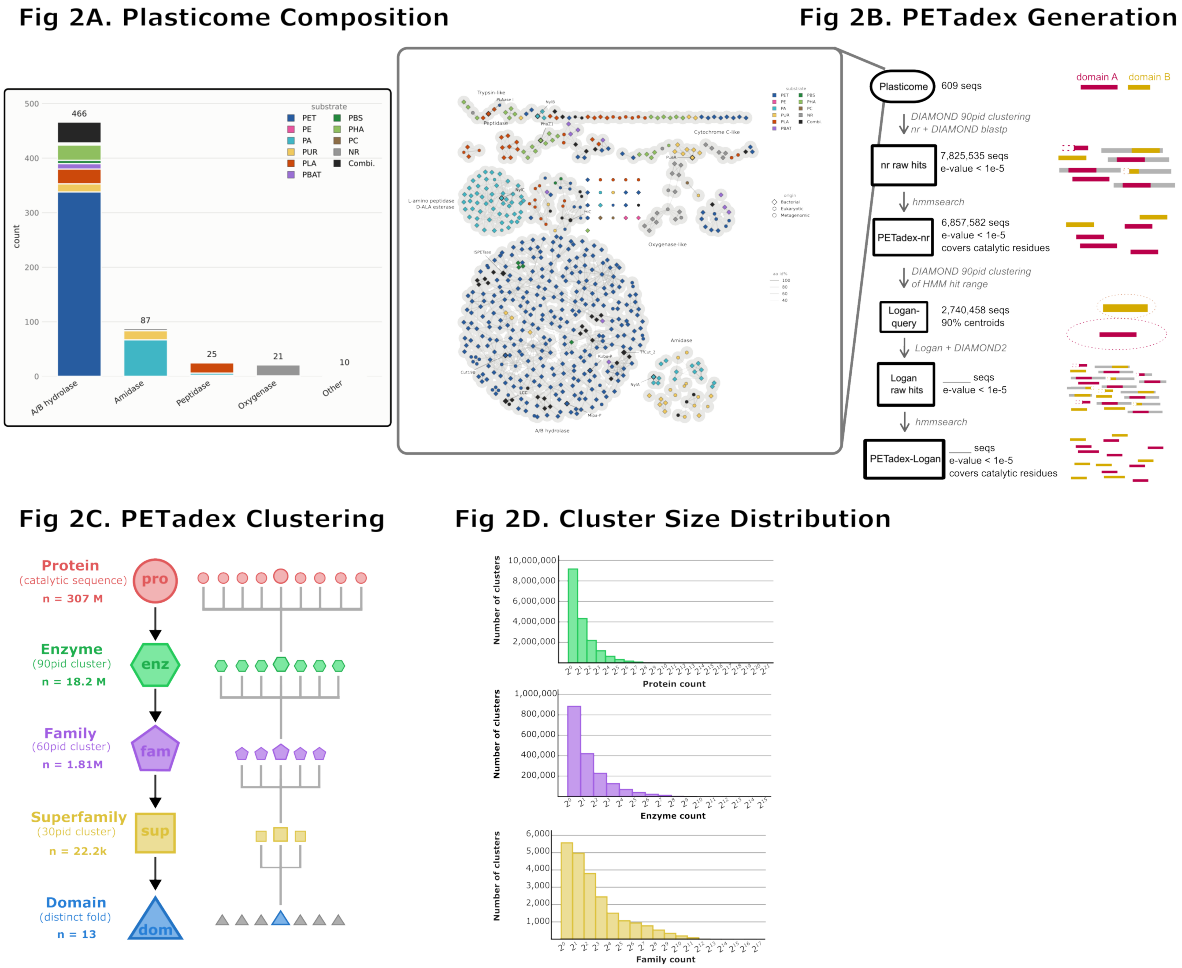
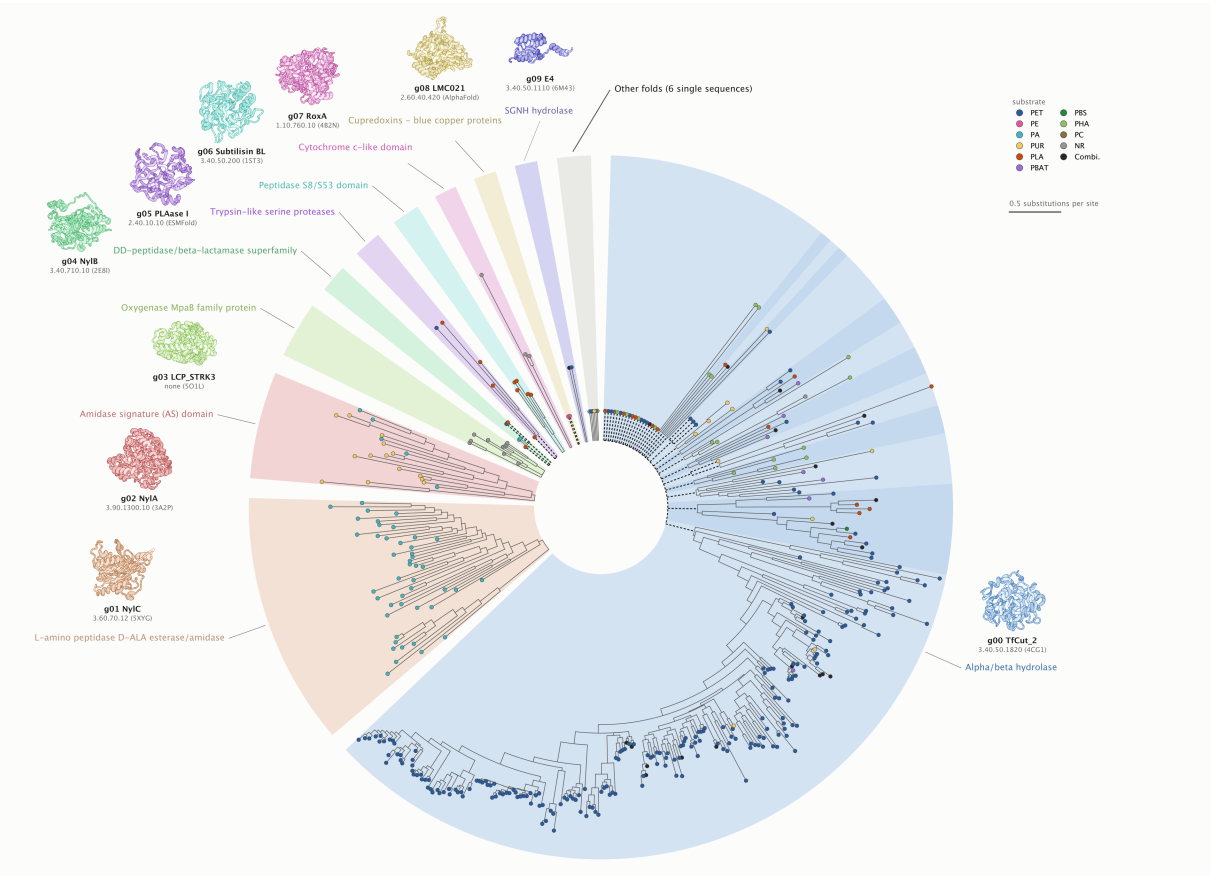


Fig. 2 The PETadex-260701 dataset. (A) Composition of the plasticome, the 609-enzyme union of the two PAZy snapshots, and its collapse to 411 cluster centroids at 90% amino acid identity, the 46 graph components those centroids resolve into, and the CATH folds assigned to them, coloured by provenance. **(B)** Overview of PETadex generation, from the 411 input sequences through the PETadex-nr and PETadex-logan expansions. **(C)** The plasticome query input, represented as a phylogenetic tree and highlighted for 16 CATH domains. **(D)** Hierarchical clustering of catalytic sequences into enzymes (90% identity), families (60%), superfamilies (30%) and distinct folds. **(E)** Cluster size distributions at the protein, enzyme and family levels.

groups based on the CATH superfamily of their catalytic domain, with membership confirmed by structural comparison, creating a stitched representation of the dataset. Phylogenetic relationships are therefore only reliable within families. Between the families of the α/β-hydrolase fold only 2 of 33 backbone branches are supported, and no relationship is inferred across folds .

Homology-based retrieval yields 1.75 billion putative plastic-degrading enzymes

To expand the dataset of known PDEs beyond the plasticome set, we first searched these 411 centroids against the NCBI nr database using DIAMOND [12], recovering 42,053,081 pairwise alignments, collapsing to 8,850,583 unique sequences. Of these, 7,825,535 pass the e-value filter ($E \leq$



1×10^{-5}), and further removing plasticome self-hits returns a final count of 7,825,107 sequences (Fig. 2B; Methods).

In order to search Logan with only the catalytic domains of these plasticome homologs, we built profile HMMs to characterize the plastic-degrading domain boundaries and catalytic residues. These HMMs were seeded from alignments of plasticome sequences together with their closest nr homologs, where each member of the alignments must contain the residues involved in the catalytic mechanism, annotated from the literature. The 46 graph components were further segmented using these catalytic mechanisms, resolving into 53 seed alignments. Each seed alignment was then expanded over four iterations of a pipeline analogous to JackHMMER [13], in which an nr homolog enters the next iteration only if it retains all of the canonical catalytic residues (Methods).

Searching these component HMMs against the 7,825,107 nr sequences yielded a significant HMM hit ($E < 1 \times 10^{-5}$) for 7,561,905 sequences. Of these, 6,764,617 had a hit whose range contained all of the positions of the catalytic residues ("complete hits"), of which 6,343,582 contained the canonical catalytic residue identities. Applied back to the plasticome, the HMMs recover 492 of its 493 non-redundant proteinsm, including 486 of them with complete hits. The ranges annotated as the catalytic domains were then clustered at 90% identity, resulting in 2,740,458 centroids (Fig. 2B; Methods). These centroids serve as the input query into Logan.

Expanding on the nr search, we queried these 2.74 million catalytic domains against Logan [Logan version. Answer is whatever the latest version is] contigs with DIAMOND blastx ($E < 1 \times 10^{-5}$), recovering 3,202,954,878 non-redundant hits (Fig. 2B). Searching these with the same component HMMs, 1,740,282,847 proteins carry

a significant HMM hit ($E < 1 \times 10^{-5}$), and 291,922,243 of them carry a complete hit spanning every catalytic position. Together with the plasticome and PETadex-nr, the dataset therefore holds 1.75 billion putative PDEs, 298.7 million of them with complete catalytic domains. α/β-hydrolases are the largest domain class in the plasticome, nr and Logan alike. We've dubbed this dataset PETadex-v2, with sub sets dubbed PETadex-plasticome (or simply plasticome), PETadex-nr and PETadex-logan, depending on the origin. [Describe a domain from the search, can be a/B hydrolase, Amydases, etc.]

Structural annotation of the PETadex enzyme space

To explore the PETadex dataset, we annotated its sequences structurally and linked each sequence to the metadata of the sample it was recovered from. We first predicted structures for the 22,235 superfamily centroids (30% amino acid identity) using ESMFold2 (Methods). These structures resolve both the catalytic domain and the accessory domains flanking it, which may additionally contribute to the catalytic mechanism.

Sample metadata provides planetary-scale annotation of the PETadex dataset

Each PETadex-logan sequence is taken from an SRA run, which links to an associated BioSample record, and potentially to an overarching BioProject. These records carry metadata features, such as the location, biome, host, etc., describing the sample, its collection conditions and subsequent sequencing methods. For our dataset of 307,155,746 ORFs, 256,236,322 are linked to BioSamples (83.4% coverage), drawn from 5,170,787 BioSamples (Fig. 2A).

Tracking the coverage of the metadata fields for the BioSamples, we can see that the BioSample records are most complete when describing collection date (47.5%), stated coordinates (32.0%) and isolation source (28.4%) (Fig. 2A). From the geographical coordinates of the samples, we can visualize the truly global span of PETadex putative PDEs, with notable hotspots in North America, Europe and East Asia. It is important to note, however, that the span of these enzymes largely tracks sequencing runs globally, or simply put, these enzymes are found anywhere and everywhere sampling occurs. Other metadata features describing the physical conditions at collection, such as temperature and pH, are rarely documented, with stated temperature available for 2.1% of ORFs and stated pH for 0.7%. This reflects the sparse and often unreliable nature of physical condition measurements throughout the SRA.

BioSample records are also highly heterogeneous, for instance the isolation source field alone contains 78,439 distinct values. To make these records comparable, and to allow enzyme candidates to be selected for experimental validation by their environment of origin, we classified BioSamples into 21 environment categories based on their stated environment terms, isolation source and host (Methods). Of the 2,744,599 BioSamples carrying at least one usable metadata field, 2,478,477 were assigned a category. Through their BioSample, 52.4% of the 307 million ORFs receive a category directly. To extend this coverage, we assigned each 90% sequence cluster the dominant category of its member ORFs and propagated it to members without a category of their own, raising coverage to 97.1% of ORFs (Fig. 2A). Counted by BioSample and by cluster, the categories give opposite pictures (Fig. 2C). Human-associated categories account for 54.3% of profiled BioSamples, predominantly *human other* (31.1%) and *human gut* (11.6%), but only 1.7% of clusters with at least two profiled ORFs. Environmental categories account for 18.7% of BioSamples but 82.2% of clusters, led by soil (31.1% of clusters) and followed by other environmental sources (16.6%), marine and freshwater (8.4% each), sediment (7.2%) and wastewater sludge (6.7%). Thus, while sequencing effort in the SRA is dominated by host-associated samples, most of the sequence diversity of PETadex enzymes lies in environmental samples.

To improve the sparse metadata annotation for environmental physical conditions of temperature (2.1%) and pH (0.7%), we attempt to leverage community species composition for BioSamples. For each run, we first averaged BacDive growth optima of the taxa detected and then weighted by read abundance to obtain an approximation of sample conditons (Methods) [14]. This proved effective at raising coverage from 2.1% to 80.6%

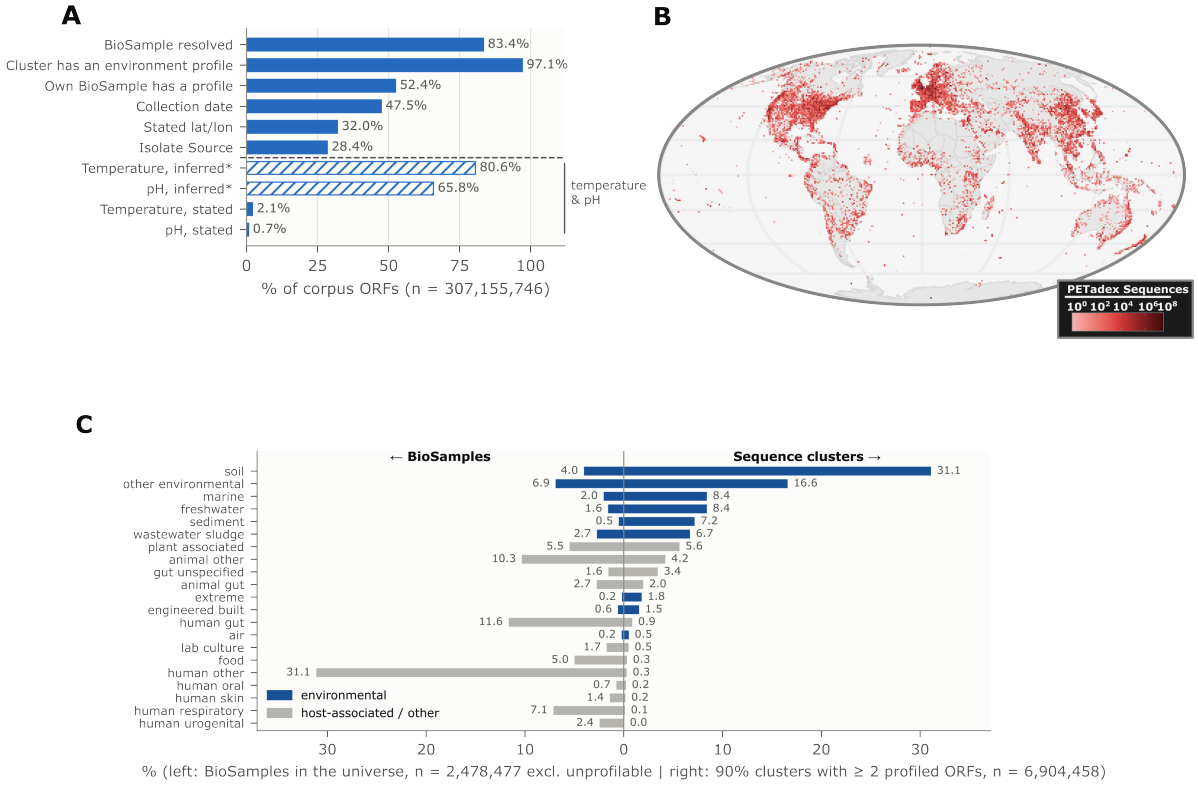


Fig. 2 Environmental metadata of the PETadex corpus. (A) Share of corpus ORFs (n = 307,155,746) with each metadata field. Solid bars, stated in BioSample records; hatched bars (*), inferred from BacDive optima of the taxa in each BioSample. **(B)** PETadex sequences by stated BioSample coordinates (log scale). **(C)** Environment categories of profiled BioSamples (left; n = 2,478,477) and of 90% clusters with $\geq$ 2 profiled ORFs (right; n = 6,904,458). Blue, environmental; grey, host-associated or other.

of ORFs for temperature, and from 0.7% to 65.8% for pH (Fig. 2A). These inferred optima describes the organism within a sample, so we validated them against stated measurements for BioSamples where data was available. Abundance-weighted optima correlate with measured pH across 6,473 BioSamples (Pearson $r = 0.34$, against 0.14 without weighting) and with measured temperature across 45,741 BioSamples ($r = 0.16$, against 0.05). These samples are compressed towards neutral pH and mesophilic temperatures, and therefore do not improve on a constant in absolute error. We therefore propose that inferred optima can be applied as a means of ranking and filtering samples, but is not reliable as an estimate of *in situ* conditions, and should thus be treated as distinct from stated values.

Together these annotations support homology-based characterization of afore uncharacterized enzymes, and allow candidates to be filtered on catalytic integrity, fold and habitat prior to testing.

PETadex is freely available online

This homology-based characterization pipeline is implemented through a DIAMOND [12] sequence search interface available freely at https://petadex.net. The sequences, catalytic domain annotations and sample metadata described here are also available for bulk download (Data availability).

Discussion

PETadex complements, rather than replaces, curated resources of plastic-degrading enzymes. PAZy [6] and PlasticDB [15] catalogue experimentally characterized enzymes, and PMBD [16] collects plastic-degrading microorganisms and genes from the literature. PETadex is several orders of magnitude larger, is derived mainly from metagenomes, and pairs each sequence with the environmental context of its source sample, while linking back to characterized enzymes through the plasticome. Apart from the plasticome, PETadex entries are putative enzymes that have not been validated experimentally. PETadex is released in numbered versions, of which the present release is PETadex-260701.

Methods

Plasticome construction

The plasticome is the union of two snapshots of the PAZy database [6]: pazy-240826 (26 August 2024; $n = 212$), the seed set of the original PETadex build [11], and pazy-260701, retrieved on 30 June 2026 (260701 is the version label of the dataset, not the retrieval date). The 495 records in the 2026 snapshot were curated manually against their source publications: 26 records without experimentally reported activity were removed and 4 sequences were added (one reported in the literature but absent from PAZy, one replacing an inactive PAZy record, and two that PAZy aggregates under a single identifier), leaving 473 enzymes. The two snapshots were joined on accession, ignoring accession version, giving a union of 609 enzymes: 397 from pazy-260701 only, 136 from pazy-240826 only and 76 present in both. Sequences were retrieved by accession, primarily from NCBI Protein.

The 609 sequences were clustered at 90% amino acid identity with USEARCH v11.0.667 [17] in a reference-seeded scheme. Thirteen well-characterized enzymes were fixed as cluster centroids, the remaining sequences were first recruited to these centroids (`usearch_global -id 0.9 -top_hit_only`), and those left unassigned were clustered *de novo* with centroids chosen by decreasing length (`cluster_fast -id 0.9 -sort length`). This produced 411 clusters (13 reference-seeded and 398 *de novo*). The 411 centroids are the query set for the NCBI-nr search. Separately, collapsing the plasticome at 100% identity gives 493 non-redundant proteins, which were used to seed the profile HMMs.

Graph components follow the definition used in Logan [10]: connected components of the all-vs-all amino acid similarity graph. The 411 centroids were aligned all-vs-all with USEARCH v11.0.667 (`-allpairs_local -acceptall`, run over both query–target orientations), and the highest-scoring alignment by bit score was kept for each pair. Pairs with ≥30% amino acid identity and $E < 1 \times 10^{-5}$ were taken as edges, and single-linkage connected components of this graph define the 46 components (25 containing a single centroid; the largest containing 232).

Domain assignment of the plasticome

Structures of all 609 plasticome enzymes were predicted with ESMFold2 (see Structure prediction). CATH homologous superfamilies were assigned by two independent routes and compared. In the structure-based route, predicted structures were segmented into domains with Merizo [18][Merizo version] and each domain was searched with Foldseek [19][Foldseek version] against the CATH S40 non-redundant domain set (CATH v4.3.0, with results confirmed against v4.4.0) [20, 21], accepting hits with TM-score ≥ 0.5 and $E \leq 1 \times 10^{-5}$. In the sequence-based route, sequences were scanned with the CATH-Gene3D HMM library (v21.0.0) and overlapping hits were resolved with cath-resolve-hits [22, 23]. Assignments were compared at the four-level CATH code, and the two routes agreed for all 643 domains matched by both. A CATH superfamily was assigned to 400 of the 411 centroids. The remaining 11 form a single component of rubber oxygenases (Pfam PF09995), a domain with no CATH superfamily in either release, and were annotated from Pfam (release 38.2) [24]. The 15 protein domains reported in Results are the distinct domain names in the catalytic residue annotation table described below, which uses the CATH superfamily name and falls back to the InterProScan [25] domain name where CATH has none.

Phylogenetic forest of the plasticome

The forest in Fig. 2C was built from the 411 plasticome centroids. Sequence-similarity families were defined as connected components of an all-vs-all similarity network of all 609 plasticome enzymes at ≥30% amino acid identity over ≥50% coverage of the shorter sequence, giving 56 families, each retaining at least one centroid. Families were grouped into fold groups by the CATH superfamily of their catalytic domain. Within each family of four or more sequences, a stratified ensemble of 16 alignments was built with Muscle 5.3 [26], the alignment with the highest mean column confidence was filtered (columns with >50% gaps or column confidence <0.5 removed), and a maximum-likelihood tree was inferred with IQ-TREE 3.1.4 [27] (ModelFinder over LG, WAG, JTT and Q.pfam; 1,000 ultrafast bootstrap replicates). Each family was represented by its most central sequence with an AlphaFold DB model (ESMFold where none existed) [28, 29]. Fold-group membership was accepted where a family reached a TM-score ≥0.5 with another member (Foldseek [19]), and links between fold groups were tested with Foldseek 3Di+AA E-values and HHalign [30] probabilities; no pair of fold groups was linked (all E > 10^{-3}, probabilities <90), and one enzyme (the PA4-degrading enzyme) whose β-lactamase-like assignment could not be confirmed was kept as a separate fold group. Within each fold group, family representatives were aligned with FoldMason [31] and a backbone tree was inferred with IQ-TREE on amino acid and 3Di partitions under a shared topology [32]. Backbone branches were kept only with ultrafast bootstrap support ≥95 and recovery by at least one of two independent trees (maximum likelihood on the amino acid partition alone, and neighbour joining on 1 − TM-score); all others were collapsed. Midpoint-rooted family trees were then grafted onto the backbone tips.

Retrieval of homologs from NCBI-nr

The NCBI nr protein database (snapshot of 25 June 2026) was exported to FASTA with `blastdbcmd` (BLAST+ 2.17.0) and formatted as a DIAMOND database (DIAMOND v2.2.4.184; 1,101,054,423 sequences, 417,051,418,494 residues) [12, 33]. The 411 plasticome centroids were searched against it with `diamond blastp --very-sensitive -k0`, the latter removing the limit on target sequences reported per query. An nr sequence hit by more than one query was assigned to its lowest-e-value query, and inherits that query's graph component. Sequences with $E \leq 1 \times 10^{-5}$ were retained. Retained sequences matching a plasticome protein by accession or by exact amino acid sequence were removed from the nr set (428), so that PETadex-plasticome and PETadex-nr are mutually non-redundant at 100% identity.

Catalytic-residue-constrained profile HMMs

For each of the 15 protein domains in the plasticome, the residues required for catalysis were annotated from the mechanistic literature and anchored to a reference sequence with a solved or predicted structure. Where cofactors make no single residue strictly essential, the conserved metal- or cofactor-ligating residues were used instead. Components whose members could not be aligned to their domain's reference with every catalytic residue in the expected column were given a component-specific reference, annotated by structural alignment to the domain reference in PyMOL.

Seed alignments were built per component from the plasticome sequences and their close nr homologs. Signal peptides were predicted and removed with SignalP 6.0 [34]. For each component, nr homologs with $E \leq 1 \times 10^{-50}$ to a plasticome query were clustered with DIAMOND `linclust` so that each component was reduced to roughly 100–1,000 sequences. Sequences were aligned with MUSCLE 5.3 (Super5 algorithm) [26], and any sequence lacking the annotated residue at a catalytic column was removed. Accessory domains and unaligned N- and C-terminal regions were then trimmed manually, and the trimmed alignments were realigned with the MUSCLE PPP algorithm. Components whose members could not all be aligned on one set of catalytic columns were split into sub-components, each with its own reference, giving 57 alignments from the 46 components. Four of these, consisting of fragments or of enzymes with no known mechanism, were discarded, leaving 53 seed alignments.

Profile HMMs were built from the seed alignments with HMMER 3.4 [35] by an iterative procedure analogous to JackHMMER [13], with the added constraint that catalytic residues are conserved. In each iteration, (i) a profile was built from the current alignment with `hmmbuild`; (ii) the reference sequence was aligned to the profile with `hmmalign`, mapping its catalytic positions onto match states; (iii) the profile was searched with `hmmsearch` against the nr homologs of its component with a DIAMOND $E < 1 \times 10^{-10}$, using the size of nr as the search space (`-Z 1101054423`); and (iv) domain hits with $E \leq 1 \times 10^{-10}$ that carried the reference residue at every catalytic match state were collected into the alignment that seeds the next iteration. If a sequence contained two domains passing these criteria, both were kept. When the next profile was built, match and insert state annotations were propagated from the previous profile, and any alignment column with an occupancy below 0.7 was declared an insert state. The procedure was run for a total of four iterations.

Each of the 53 profiles was searched with `hmmsearch` against the nr sequences assigned to its own component, out of the 7,825,107 retained, and the reference catalytic positions were transferred to each hit. A sequence therefore belongs to one component, that of its best DIAMOND query, and can match more than one profile only where a component has sub-component profiles; one record was kept per sequence, its hit with the lowest E-value. A hit was called complete if it has a residue at every catalytic position, and canonical if, in addition, each of those residues matches the reference. The catalytic domain of a sequence was defined as the region of the sequence aligned to the profile. Sequences with only incomplete hits were kept in PETadex-nr and flagged (797,288 of 7,561,905), but were excluded from the Logan query. The catalytic domains of complete hits were clustered within each component at 90% identity (`diamond linclust --id 90 --ext global --comp-based-stats 0`, DIAMOND v2.2.4.184, default coverage) to give the Logan query.

Logan search and hierarchical clustering

The Logan query (2,740,458 catalytic domains) was searched against Logan contigs [10][confirm the Logan version and the number of accessions searched] with DIAMOND `blastx` at $E < 1 \times 10^{-5}$ [DIAMOND version, sensitivity mode, any identity or coverage filters], giving 3,202,954,878 non-redundant hits [how hits were turned into protein sequences: ORF calling tool, minimum length, treatment of ORFs truncated at contig ends]. The 53 profile HMMs were searched against these sequences as for nr, with catalytic domains and catalytic residues annotated in the same way. Sequences with a significant hit ($E < 1 \times 10^{-5}$; 1,740,282,847) form PETadex-logan, and, as for nr, those whose hits are incomplete are kept and flagged (1,448,360,604 of them; 291,922,243 have a complete hit).

Catalytic domains from the plasticome, nr and Logan were clustered with DIAMOND at 90%, 60% and 30% amino acid identity, which we term enzymes, families and superfamilies [clustering command, coverage thresholds, DIAMOND version]. Clustering was hierarchical, in that the centroids of one level were the input to the next, and ordered by origin, so that centroids are drawn preferentially from characterized sequences. Plasticome domains were clustered first. nr domains were assigned to a plasticome centroid where they aligned to one at the given threshold, and the remainder were clustered among themselves. Logan domains were assigned to plasticome or nr centroids in the same way before the remainder were clustered. Full-length sequences were then joined back to each catalytic domain. [state how folds were assigned to superfamilies (Fig. 2D)]

Structure prediction

Structures were predicted with ESMFold2 (`biohub/ESMFold2`, release `esmfold2-2026-05`), which uses the ESMC-6B language model as its trunk, at bfloat16 precision with a fixed random seed on NVIDIA A100 GPUs [number of recycles; not recorded]. For the 609 plasticome enzymes the mean pLDDT was 0.94, and 605 structures (99.3%) had a mean pLDDT of at least 0.70. [Unconfirmed that the 22,235 superfamily centroids were folded with ESMFold2: the other

large folding run on record used ESM3-open, and the stored ESMFold2 structures are labelled as the 60% level. Confirm the model and the clustering level, and give the pLDDT summary and any sequence-length limit.]

BioSample metadata and environment categories

Each Logan ORF carries the accession of the SRA run it was assembled from. Runs were linked to BioSample and BioProject records through SRA metadata [source and retrieval date of the SRA and BioSample metadata]. Of the 7,626,110 runs contributing ORFs, 1,414,327 have no BioSample record, which accounts for the ORFs without one. The map in Fig. 2B uses only coordinates stated in BioSample records.

BioSamples were classified into 21 environment categories by rule. Nine are environmental (soil, marine, freshwater, sediment, wastewater sludge, extreme, engineered or built, air and other environmental) and twelve are host-associated or other (human gut, human oral, human skin, human respiratory, human urogenital, human other, animal gut, animal other, gut unspecified, plant-associated, laboratory culture and food). Temperature, pH, salinity and depth were not used for classification. Of 6,796,722 BioSamples, 2,744,599 carried at least one usable field, and 2,478,477 of these were assigned a category.

Each 90% cluster was assigned its dominant category, the most frequent category among member ORFs whose BioSample was classified. An ORF with a classified BioSample keeps its own category; an ORF without one takes the dominant category of its cluster, and is therefore annotated by inference. Clusters with a single classified ORF were excluded from cluster-level comparisons of categories (Fig. 2C), because a single observation assigns a cluster wholly to one category by construction; this leaves 6,904,458 of the 14,455,359 clusters with a dominant category. Shares of BioSamples per category are not corrected for the size of individual BioProjects [decide whether to cap BioSamples per BioProject and category; capping at 25 moves human other from 28.9% to 17.0%].

Inferred temperature and pH

The temperature and pH of each BioSample were inferred from the organisms detected in it. Read counts per taxon for each run were taken from the SRA taxonomic analysis [36], counting only reads assigned directly to a taxon and not those assigned to its descendants, and were summed per taxon over the runs of a BioSample so that each taxon contributes once. Taxon identifiers were resolved to current NCBI taxonomy and matched to growth optima in BacDive [14][the values in Results use an earlier BacDive cache and predate the full retrieval of 8 September 2026 (102,187 strains); recompute and give that date], excluding entries above species rank and non-prokaryotes (612 taxa with a temperature optimum, 203 with a pH optimum). The inferred value is the mean of the optima of the taxa detected, weighted by their read counts, and is undefined where no taxon with an optimum has reads. Inferred values were compared with the temperature and pH stated in BioSample records where both exist (45,741 and 6,473 BioSamples) by Pearson correlation, for the read-weighted mean and for an unweighted mean over taxa.

Statistics

Agreement between inferred and stated temperature and pH was measured with the Pearson correlation coefficient.

Supplementary information.

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Declarations

- Funding
- Competing interests
- Ethics approval and consent to participate
- Consent for publication
- Data availability: PETadex sequences, annotations and metadata are available at https://petadex.net.
- Materials availability
- Code availability: the website source code is available at https://github.com/petadex/petadex.io.
- Author contribution

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